Experiment No.	3
Experiment Title	https://github.com/sugith-2025/MachineLearning/blob/main/Experimen
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1 Objective

To build and compare Linear Regression with its regularized variants – Ridge, Lasso and Elastic Net – for predicting the loan sanction amount of an applicant, and to study how the regularization strength (α) affects model coefficients, training error and validation error.

2 Problem Statement

Given an applicant's demographic, financial and property-related attributes (income, credit score, requested loan amount, employment type, property value, etc.), the task is to predict a continuous target, *Loan Sanction Amount (USD)*, that a lending institution is likely to approve. This is a supervised regression problem. The input is a tabular record of 23 applicant features, and the output is a single real-valued prediction of the sanctioned amount. The objective is to identify which regression formulation generalizes best and how penalizing large coefficients changes model behaviour.

3 Dataset Description

Dataset Name	Loan Sanction Amount Prediction (`train.csv`)
Dataset Source	Kaggle – Loan Sanction Amount Prediction dataset
Number of Samples	30,000 (raw); 29,660 after removing rows with a missing target
Number of Features	23 raw input features, expanded to 45 after one-hot encoding of categorical columns
Number of Classes	Not applicable – the target is continuous (regression task)
Missing Values	Present in *Gender* (53), *Income* (4576), *Income Stability* (1683), *Type of Employment* (7270), *Current Loan Expenses* (172), *Dependents* (2493), *Credit Score* (1703), *Has Active Credit Card* (1566), *Property Age* (4850), *Property Location* (356) and the target *Loan Sanction Amount* (340)
Train-Test Split	80% train (23,728 samples) / 20% test (5,932 samples), `random_state=42`

4 Exploratory Data Analysis

Before modelling, the target distribution and its relationship with two of the strongest expected predictors – income and credit score – were examined.

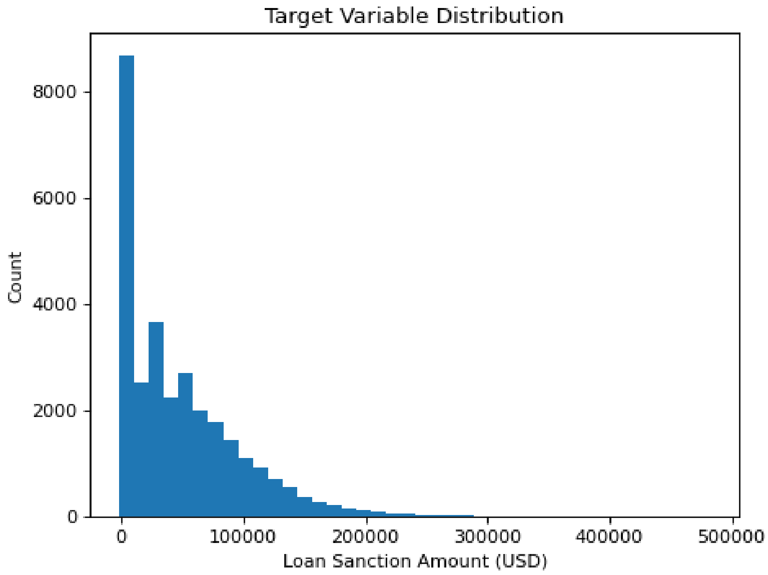

Figure 1: Distribution of the target variable, Loan Sanction Amount (USD).

Inference: The sanctioned loan amount is heavily right-skewed: most applicants are sanctioned relatively small amounts, while a long tail extends toward much larger sanctions. This skew suggests that a plain linear model may struggle at the extremes and that error metrics such as MAE will be dominated by the dense low-value region. It also hints that a log-transform of the target could be explored in future iterations, although it was not applied in this experiment to keep the pipeline consistent with the notebook.

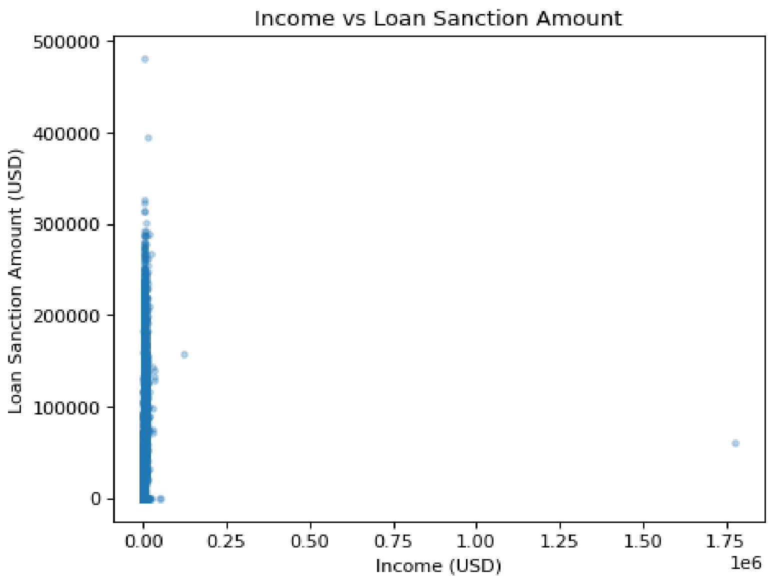

Figure 2: Income (USD) vs. Loan Sanction Amount (USD).

Inference: Most applicants are concentrated at low-to-moderate income levels, with a visible but noisy upward trend in sanctioned amount as income rises. A handful of high-income applicants pull the axis outward without a proportionally large increase in sanction amount, indicating that income alone is a weak and non-linear predictor and that other factors (loan request size, credit history) likely dominate the final decision.

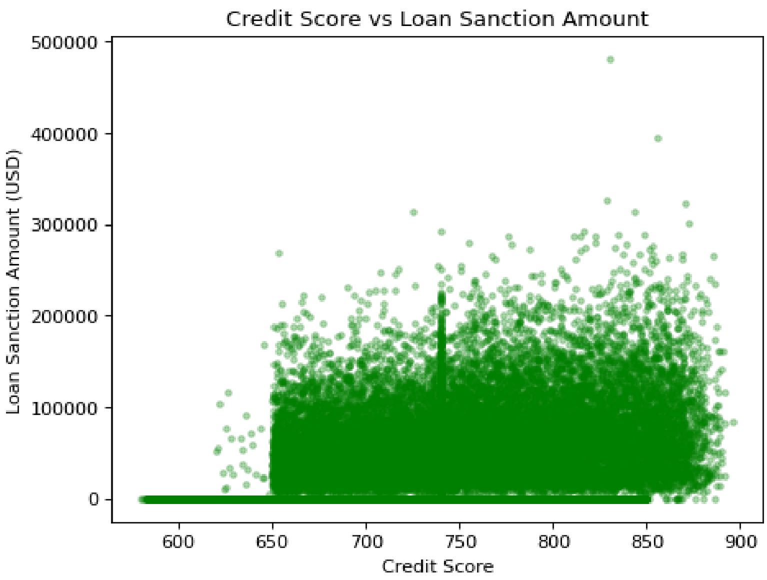

Figure 3: Credit Score vs. Loan Sanction Amount (USD).

Inference: Credit score, ranging roughly between 500 and 900, shows a mild positive association with the sanctioned amount, but the spread at every score value is large. This suggests credit score contributes meaningfully but is not sufficient on its own to explain the target, reinforcing the need for a multivariate model rather than a single-feature rule.

5 Data Preprocessing

- **Column removal:** *Customer ID*, *Name* and *Property ID* were dropped as they are identifiers with no predictive value, reducing the frame from 24 to 21 columns.
- **Target-row removal:** 340 rows with a missing *Loan Sanction Amount* were dropped, since the target cannot be imputed without introducing bias, leaving 29,660 rows.
- **Missing value handling:** The 11 numeric columns were imputed with their column median (robust to the skew seen in Section 4), and the 9 categorical columns were imputed with their column mode.
- **Encoding:** All categorical columns were one-hot encoded using `pd.get_dummies(drop_first=Tru` expanding the frame to 46 columns (45 features + target).
- **Feature scaling:** All features were standardized using `StandardScaler` (zero mean, unit variance), fit only on the training split and applied to both splits, which is essential for Ridge/Lasso/Elastic Net since their penalty terms are scale-sensitive.

- **Train-test split:** An 80/20 split was used (`train_test_split`, `random_state=42`), giving 23,728 training and 5,932 test samples.

6 Machine Learning Algorithm

Four linear models were trained on the same standardized feature matrix $X \in \mathbb{R}^{n \times p}$ against target y:

- **Ordinary Least Squares (baseline):**

$$\min_w \|y - Xw\|_2^2$$

  No penalty is applied to the coefficients, so the model can overfit when features are correlated or numerous.

- **Ridge Regression (L_2 penalty):**

$$\min_w \|y - Xw\|_2^2 + \alpha \|w\|_2^2$$

  Shrinks all coefficients toward zero smoothly, which is effective against multicollinearity but never eliminates a feature entirely.

- **Lasso Regression (L_1 penalty):**

$$\min_w \frac{1}{2n} \|y - Xw\|_2^2 + \alpha \|w\|_1$$

  Drives some coefficients exactly to zero, performing implicit feature selection.

- **Elastic Net ($L_1 + L_2$ penalty):**

$$\min_w \frac{1}{2n} \|y - Xw\|_2^2 + \alpha\, r \|w\|_1 + \frac{\alpha(1-r)}{2} \|w\|_2^2$$

  where r is the `l1_ratio`. It combines Ridge's stability with Lasso's sparsity.

Advantages: Regularized regressors are fast to train, easy to interpret via their coefficients, and resistant to overfitting on high-dimensional, correlated tabular data such as this one-hot encoded loan dataset.

Limitations: All four models are linear in the input features, so they cannot capture non-linear interactions (e.g. income and credit score jointly influencing sanction amount) without manual feature engineering. Lasso and Elastic Net also require iterative solvers that can be sensitive to the convergence tolerance and maximum iteration count on large feature sets.

7 Experimental Setup

Python Version	3.11
Scikit-Learn Version	1.x (as imported in the notebook kernel)
IDE	Jupyter Notebook
Operating System	Notebook execution environment (kernel: `python3`)
Hardware	Standard CPU (no GPU required for linear models)

8 Hyperparameter Selection

A 5-fold `GridSearchCV` (scoring = R^2) was used to tune each regularized model on the training split.

Parameter	Value Searched
Ridge α	{0.01, 0.1, 1, 10, 100}
Lasso α	{0.001, 0.01, 0.1, 1, 10}
Elastic Net α	{0.01, 0.1, 1, 10}
Elastic Net `l1_ratio`	{0.2, 0.5, 0.8}

Best parameters found:

Model	Best Parameters	Best CV R^2
Ridge	$\alpha = 100$	0.574
Lasso	$\alpha = 10$	0.570
Elastic Net	$\alpha = 0.1$, `l1_ratio` = 0.5	0.584

Elastic Net achieved the best cross-validated R^2 among the tuned models, suggesting that a balanced mix of L_1 and L_2 penalty helps more than either extreme alone on this feature set.

9 Results

Final models were refit on the full training split using their best parameters. Since this is a regression task, standard regression metrics are reported instead of classification metrics.

5-Fold Cross-Validation Performance (on training data):

Model	MAE	MSE	RMSE	R^2
Linear	21549.688	1.011×10^9	31798.226	0.568
Ridge	21566.836	9.961×10^8	31560.930	0.574
Lasso	21543.656	1.005×10^9	31707.011	0.570
Elastic Net	21743.082	9.727×10^8	31188.079	0.584

Held-out Test Set Performance:

Model	MAE	MSE	RMSE	R^2
Linear	21592.264	1.019×10^9	31922.006	0.551
Ridge	21589.231	1.018×10^9	31902.603	0.552
Lasso	21569.424	1.018×10^9	31912.953	0.551
Elastic Net	21755.244	1.019×10^9	31927.292	0.551

On the held-out test set the four models perform almost identically ($R^2 \approx 0.55$), even though Elastic Net had the strongest cross-validated score during tuning – indicating that regularization mainly stabilizes the coefficients rather than dramatically changing predictive accuracy on this dataset.

10 Result Visualization

Figure 4: Test set R^2 comparison across Linear, Ridge, Lasso and Elastic Net.

Inference: The four bars sit almost at the same height ($R^2 \approx 0.55$), confirming that the choice of regularizer has very little effect on final test-set accuracy for this dataset. This is common when the underlying signal-to-noise ratio in the data is the true bottleneck rather than overfitting – regularization cannot recover information the raw features do not contain.

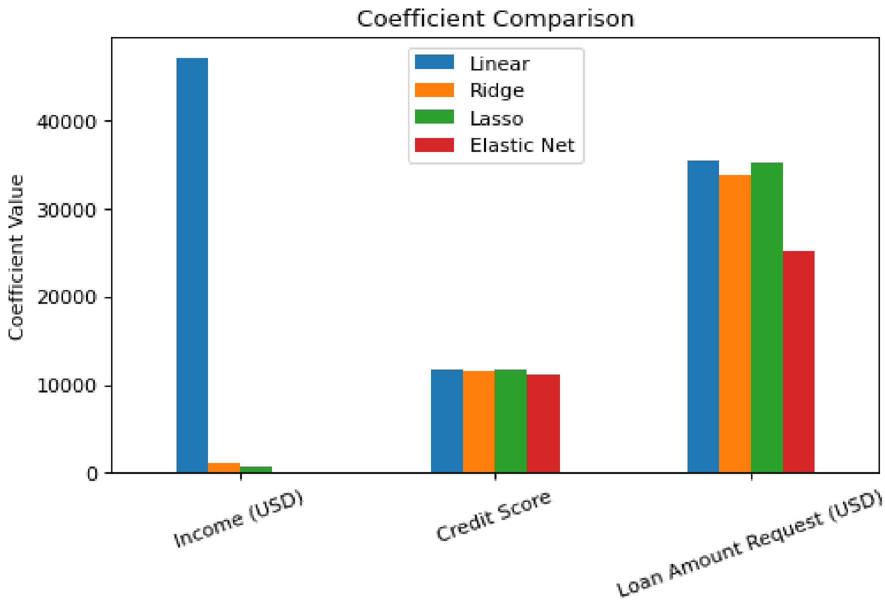

Figure 5: Coefficient comparison for Income, Credit Score and Loan Amount Request across the four models.

Inference: The plain Linear Regression coefficient for *Income* is extreme (≈47,114) compared to Ridge (≈1026), Lasso (≈735) and Elastic Net (≈93), showing that OLS was assigning an unstable, inflated weight to a feature that is likely correlated with others. All three regularized models pull this coefficient sharply toward zero. In contrast, the *Credit Score* coefficient stays close to 11,200–11,640 across all four models, indicating it carries a genuinely stable, non-redundant signal that regularization does not need to suppress. *Loan Amount Request* shrinks moderately, most under Elastic Net, consistent with its dual penalty.

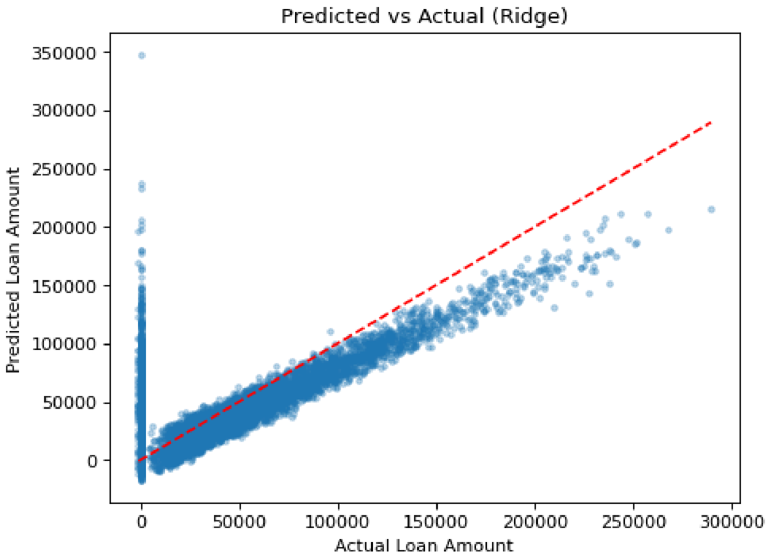

Figure 6: Predicted vs. Actual Loan Sanction Amount (Ridge model).

Inference: Points cluster around the ideal $y = x$ line for low-to-moderate loan amounts but fan out and increasingly fall below the line as the actual amount grows, meaning the model systematically under-predicts large sanctions. This mirrors the right-skewed target distribution seen in Section 4 and confirms that a purely linear fit under-serves the tail of the distribution.

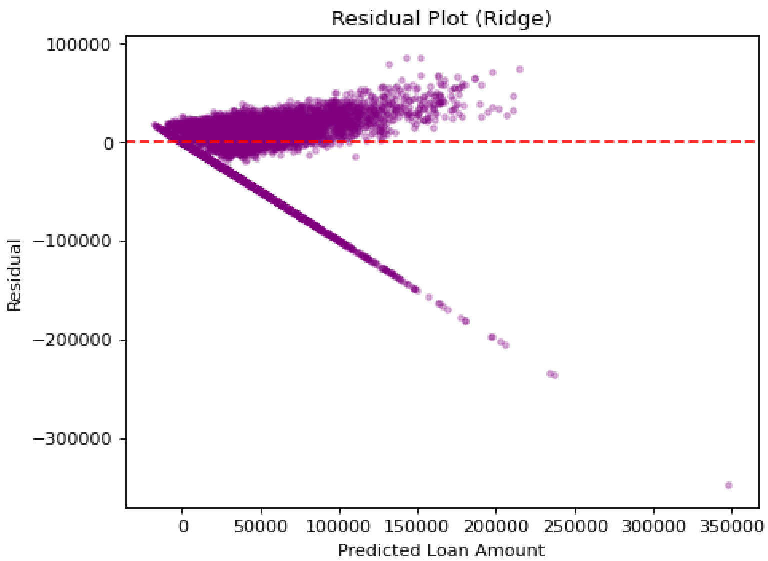

Figure 7: Residual Plot for the Ridge model (Predicted Value vs. Residual).

Inference: The residuals fan outward as the predicted value increases – a funnel or cone shape – which is a classic sign of heteroscedasticity: prediction error variance is not constant across the range of the target. While residuals are roughly centered at zero (no strong systematic bias), the growing spread at higher predicted amounts means confidence in individual predictions should decrease for high-value loans.

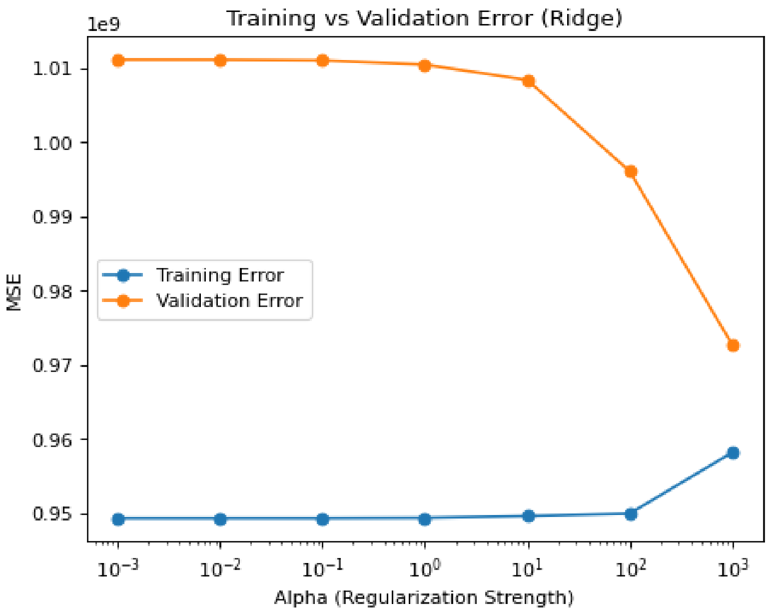

Figure 8: Training vs. Validation Error across Ridge regularization strength α (log scale).

Inference: For small to moderate α (0.001 to around 10), training and validation MSE remain low and close together, showing the model is neither over- nor badly underfitting in that range. Beyond $\alpha \approx 100$, both errors rise sharply and converge upward, which is classic underfitting – the penalty has become strong enough to force most coefficients toward zero, discarding useful signal along with noise. This confirms why the grid search selected a value near this boundary ($\alpha = 100$) rather than a much larger one.

11 Discussion

Across all four linear formulations, test-set R^2 plateaus around 0.55, meaning roughly half of the variance in loan sanction amount is explained by the available features under a linear assumption. Regularization did not materially change accuracy here, but it delivered two other benefits that matter in practice: it stabilized the wildly inflated *Income* coefficient produced by plain OLS, and it gave a cleaner, more trustworthy picture of feature importance (*Credit Score* and *Loan Amount Request* emerged as the consistently influential predictors). The residual and predicted-vs-actual plots both point to the same limitation – error grows with the size of the sanctioned amount – suggesting that either a log-transformed target or a non-linear model (e.g. gradient boosting) would likely outperform these linear baselines on the upper end of the distribution. A natural next step would be to compare these results against a tree-based ensemble to quantify how much of the remaining 45% unexplained variance is due to genuine non-linearity versus dataset noise.

12 Conclusion

This experiment implemented and compared Linear, Ridge, Lasso and Elastic Net regression for predicting loan sanction amounts on a 29,660-row applicant dataset. After hyperparameter tuning via 5-fold cross-validation, Elastic Net ($\alpha = 0.1$, `l1_ratio`$= 0.5$) achieved the best cross-validated R^2 (0.584), while all four models converged to a similar test R^2 of about 0.55. The key takeaway is that regularization here served mainly as a stabilizer of coefficient estimates – most visibly shrinking the unstable *Income* weight – rather than a lever for large accuracy gains, and that the residual heteroscedasticity observed points toward non-linear modelling as the more promising direction for future improvement.

13 References

https://github.com/sugith-2025/MachineLearning/blob/main/Experiment_3.ipynb.